

Constructor University

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Lab Experiment 5: Filters

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1. Introduction

Filter are Electrical circuits which allows us to select specific range of frequencies and pass them and block others. There are two main types of filters passive and active. Passive filters consist of resistors capacitors and Inductors and the gain of certain frequency can't be more than 1. On the other hand, active filters can have OpAmps or transistors, it amplifies the certain frequencies.

In this experiment, we observed passive filters. Main focus was observing output voltage and phase shift change while the frequency was changed. The frequency response of the filters was then analyzed using Bode plots.

The most important quantities are Voltage ration, the attenuation in decibels, the phase shift, and the cut-off frequency.

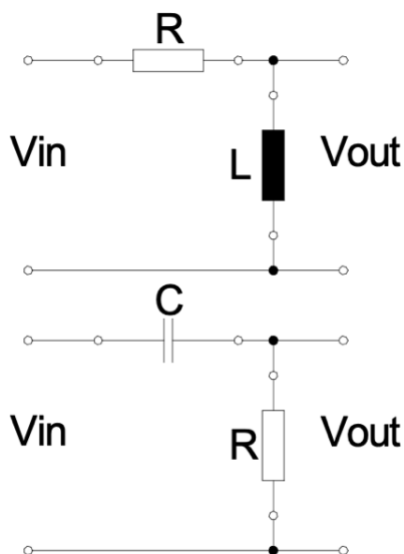
Voltage transfer ratio is given by: $A = V_{out}/V_{in}$

Attenuation in decibels is given by: $A_{db} = 20 \log \left[(V_{out}/V_{in}) \right]$

The cut-off frequency is the frequency at which the output voltage drops to : $1/\sqrt{2}$ of its max value.

1.1 High-Pass Filter

A high-pass filter allows high-frequency signals to pass while blocking the other ones. In a passive RC high-pass filter, the capacitor is placed in series with the input signal and the resistor is connected to ground. The output voltage is measured across the resistor. At very low frequencies, the capacitor behaves is like an open circuit, so only a very small output voltage is measured. At high frequencies, the capacitor behaves like a short circuit, so most of the input voltage appears across the resistor.



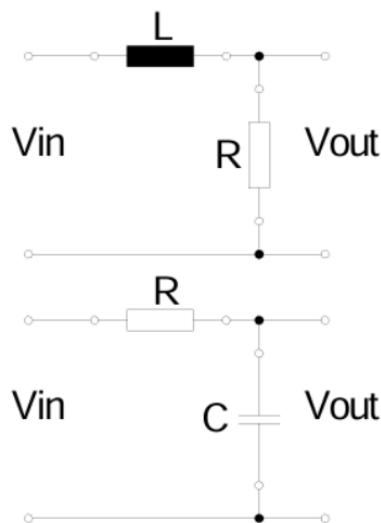
The magnitude and the phase of RC transfer function is calculated by this equation:

$$|A| = \frac{1}{\sqrt{1 + \frac{1}{(wRC)^2}}} \quad \phi = \arctan\left(\frac{1}{wRC}\right). \text{ where } w = 2\pi f$$

And the theoretical cut off frequency is $f_c = \frac{1}{2\pi RC}$

1.2 Low-Pass Filter

A low-pass filter has the opposite behavior of a high-pass filter. It allows low-frequency signals to pass while attenuating high-frequency signals. In an RC low-pass filter, the resistor is placed in series with the input signal and the capacitor is connected to ground. The output voltage is measured across the capacitor.

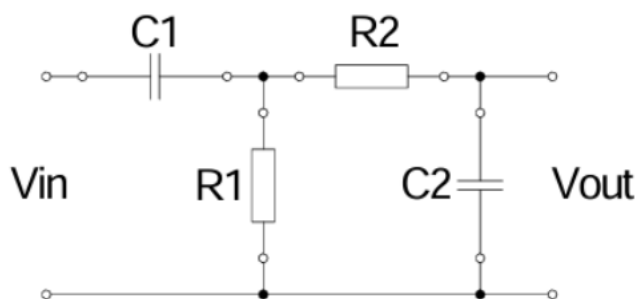


At low frequencies, the capacitor has a high impedance, so most of the input voltage appears across it. At high frequencies, the capacitor behaves like a short circuit, so the output voltage decreases.

The magnitude of the transfer function of a low-pass filter is: $|A| = \frac{1}{\sqrt{1 + \frac{1}{(wRC)^2}}}$ and the phase shift is $\phi = -\arctan(wRC)$ and cut-off frequency is: $f_c = \frac{1}{2\pi RC}$

1.3 Band-Pass Filter

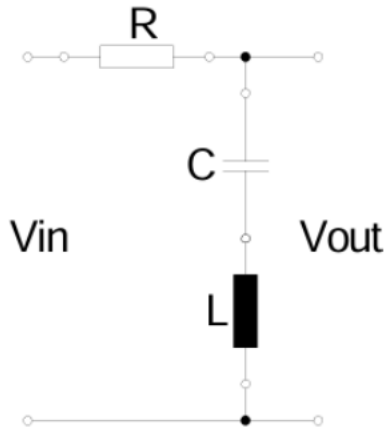
Band pass filter is only allowing certain frequency range and blocking everything below and above it. It combines the high-pass and low-pass filter.



The most important quantities for a band-pass filter are the lower cut-off frequency, the upper cut-off frequency, the bandwidth, and the center frequency. The bandwidth is calculated as $B = f_2 - f_1$ and the center frequency is $f_0 = \sqrt{f_1 f_2}$. f_0 is frequency where output is max

1.4 Notch Filter

Notch filter is opposite of the band-stop filter. It blocks certain range of frequencies and allow everything above and below.



For notch the center frequency is calculated by: $f_0 = \frac{1}{2\pi\sqrt{LC}}$. Here output reaches min value.

The cutoff values are obtained by this equations $f_{+,-} = \frac{\pm RC \pm \sqrt{(RC)^2 + 4LC}}{2\pi LC}$

1.5 Bode and Nyquist Plots

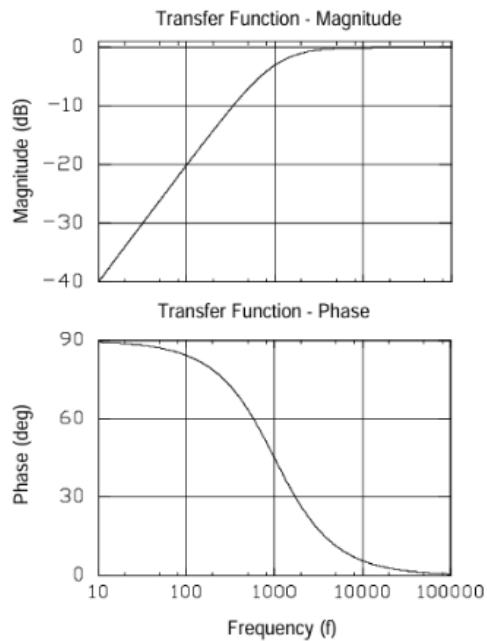
Bode plot or Nyquist plots are ways to visualize magnitude and frequency phase response of the filter. A Bode plot consists of two graphs. The first graph shows the magnitude of the transfer function in decibels as a function of frequency, usually on a logarithmic scale. The second graph shows the phase shift as a function of frequency.

The magnitude in decibels are calculated using: $A_{db} = 20 \log\left(\frac{V_{out}}{V_{in}}\right)$

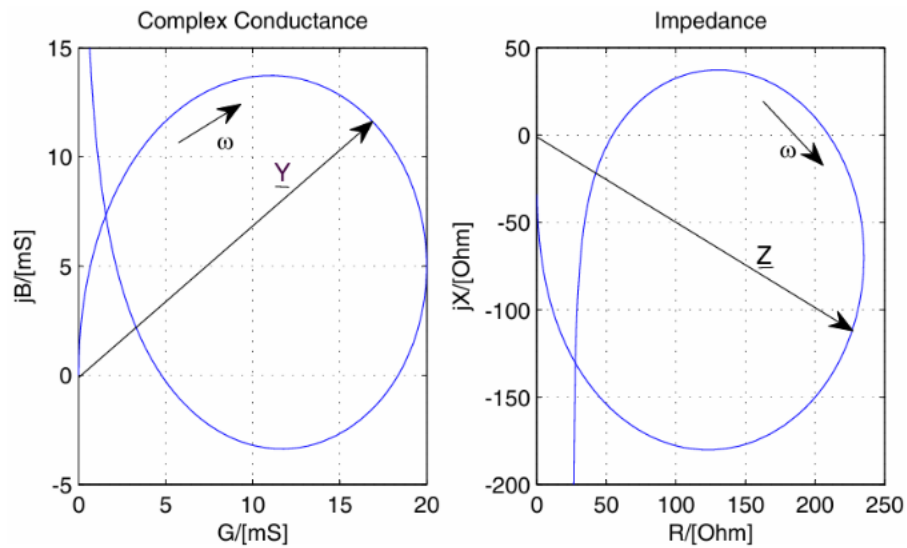
In a Nyquist plot, the real and imaginary parts of the transfer function are plotted against each other.

The real part is calculated as $Re(A) = \frac{V_{out}}{V_{in}} \cos \phi$ and imaginary part: $Im(A) = \frac{V_{out}}{V_{in}} \sin \phi$

This is example of bode-plot



This is Nyquist plot

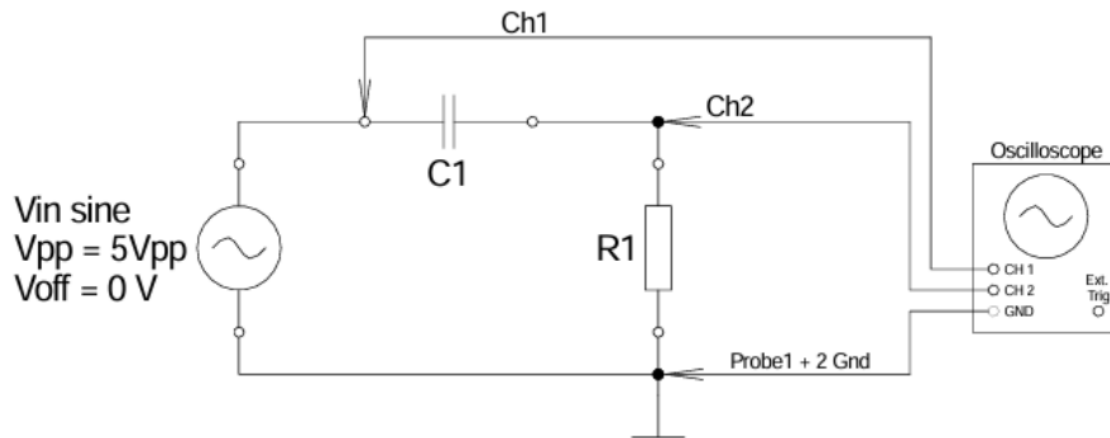


2. Execution

2.1 Part 1: High-pass filter-Setup

The objective of this experiment was to investigate the characteristics of high-pass filter. The circuit was built using a capacitor and a resistor. The input signal was supplied by a wave generator, and the input and output voltages were measured using an oscilloscope.

Channel 1 of the oscilloscope was connected to the input voltage, while Channel 2 was connected to the output voltage across the resistor. The function generator was set to produce a sinusoidal signal with no DC offset.



The component values used were $C=1.5\text{nF}$ $R=22\text{K ohm}$

The theoretical cut-off frequency was $f_c = \frac{1}{2\pi RC}$, with the values plugged we get $f_c = \frac{1}{2\pi(22000)(1.5 \times 10^{-9})} \approx 4822.9\text{Hz}$

2.2 High-Pass Filter-Execution and Results

The frequency was varied from low values to high values. For each frequency, the peak-to-peak input voltage, peak-to-peak output voltage, and phase shift were recorded. The purpose was to observe how the output amplitude increased as the frequency increased.

Part 1 : Hi-Pass			
Frequency (Hz)	Input Amplitude (V)	Output Amplitude (V)	Phase (degree) *reference ch2-ch1
50	5.4V	172mV	-87
100	5.4V	328mV	-84
200	5.4V	660mV	-81.3
500	5.4V	1.52V	-71.4
1k	5.4V	2.72V	-55.5
2k	5.4V	4.2V	-34
5k	5.4V	4.8V	-14.4
10k	5.4V	5V	-6.5
20k	5.4V	5V	-3.6
50k	5.4V	5V	-1.19

100k	5.4V	5V	2.16
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2.3 Notch Filter- Setup

Aim of the second part was to investigate the behavior of notch filter. The circuit was built using a resistor, capacitor, and inductor. The function generator supplied a sinusoidal input signal, and the oscilloscope was used to measure the input and output voltage and the phase shift. The component values were: R=1.5 k ohm, C=15 nF and L=10 mH the theoretical center frequency was calculated using this values:

2.4 Notch Filter-Execution and Result

The theoretical values was calculated using given values. The central frequency was calculated

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi(10 \times 10^{-3})(15 \times 10^{-9})} \approx 12995 \text{ Hz} \quad f_0 = 12995 \text{ Hz}$$

The theoretical cutoff frequencis were: $f_{\pm} = \frac{\pm 1500 \times 15 \times 10^{-9} + \sqrt{(1500 \times 15 \times 10^{-9})^2 + 4(10 \times 10^{-3} \times 15 \times 10^{-9})}}{2\pi \times 2(10 \times 10^{-3} \times 15 \times 10^{-9})}$

$$f_+ \approx 29582 \text{ Hz} \quad \text{and} \quad f_- \approx 5709 \text{ Hz}$$

$$\text{There for the bandwidth is } B = f_+ - f_- = 29582 - 5709 = 23873 \text{ Hz}$$

The output voltage decreased as the frequency approached the center frequency and reached a minimum near the resonance frequency. After this point, the output voltage increased again. This confirms the expected behavior of a notch filter.

Part 2 : Notch			
Frequency (Hz)	Input Amplitude (V)	Output Amplitude (V)	Phase (degree) *CH2 -CH1
1k	5.4	5.2	7.94
2k	5.4	5.2	13.7
5k	5.4	4	30.5
5.71k	5.4	3.52	41.9
7k	5.4	2.96	51.8
9k	5.4	1.88	64.6
11k	5.4	900mV	68.9
12.91k	5.4	208mV	-1.84
13k	5.4	216mV	-14.4
15k	5.4	860mV	-66
18k	5.4	1.76	-62.8

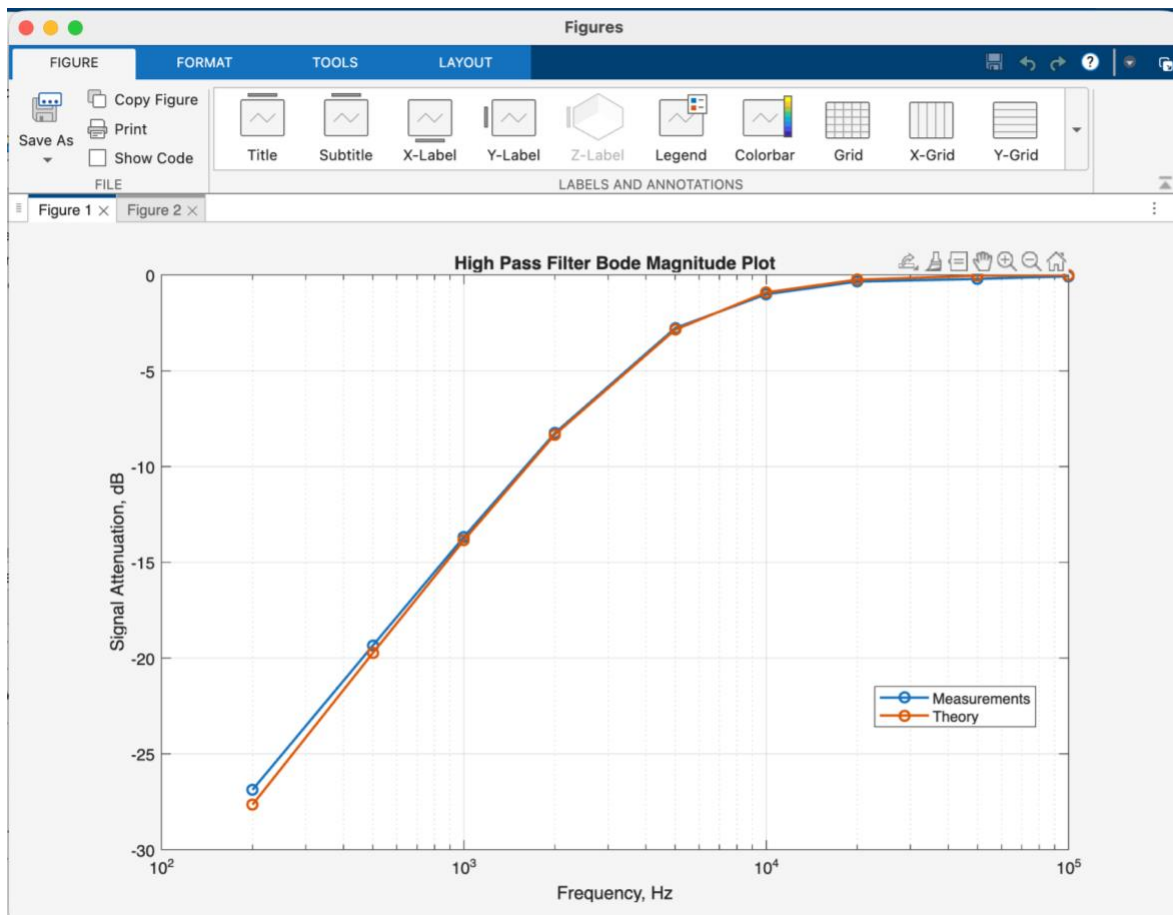
22k	5.4	2.64	-52.2
26k	5.4	3.2	-46.8
29.58k	5.4	3.6	-39.2
35k	5.4	4	-40.7
50k	5.4	4.6	-21.5
100k	5.4	5.2	-10.7

3. Evaluation

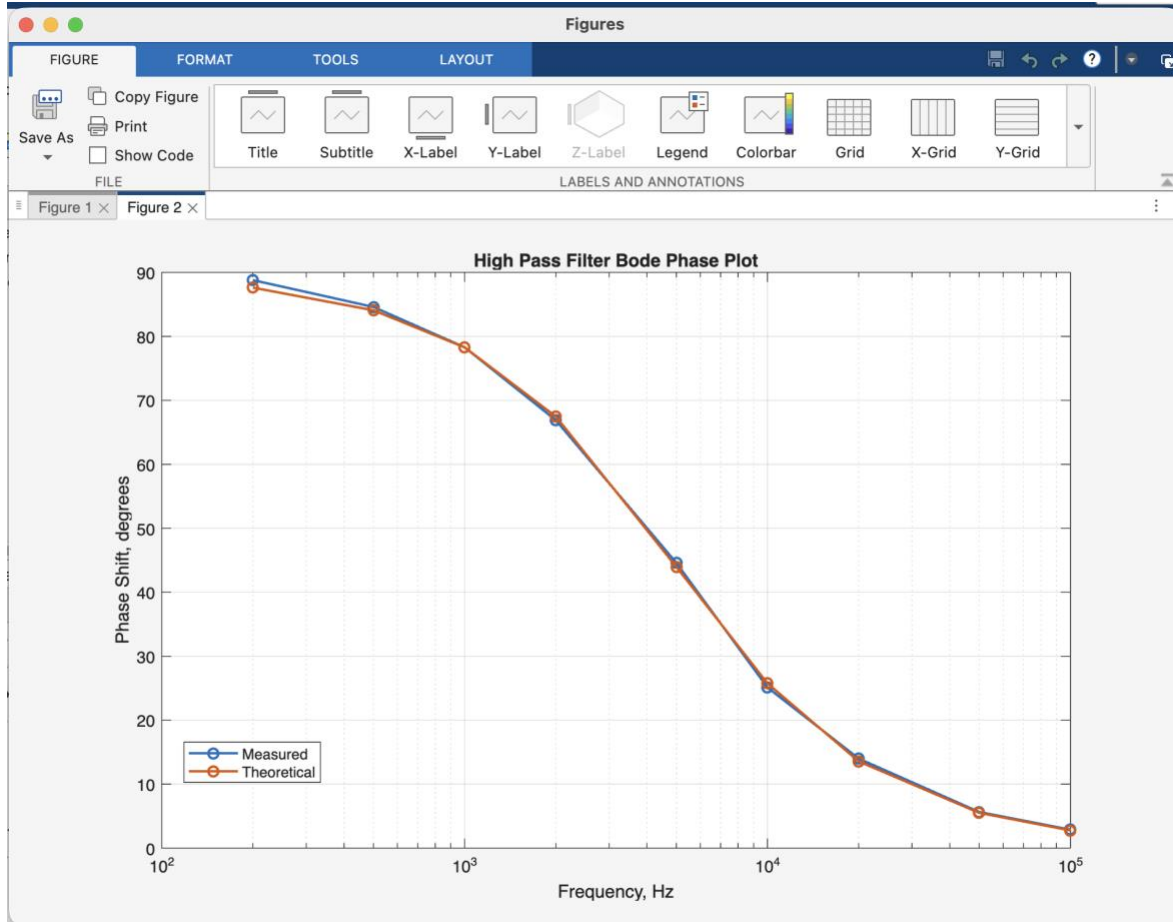
3.1 High-pass Filter-Evaluation

The high-pass filter was evaluated by comparing the measured output voltage and phase shifts to the theoretical behavior of an RC high-pass filter

High-Pass filter bode magnitude plot



High-Pass filter bode phase plot



Using the measured values from the high-pass filter, the magnitude and phase behavior can be compared with the theoretical high-pass response. The measured attenuation was calculated using this equation $A_{db} = 20 \log \left(\frac{V_{out}}{V_{in}} \right)$ and theoretical magnitude was calculated using

$$A = \frac{1}{\sqrt{1 + \frac{1}{(wRC)^2}}} \text{ where: } w = 2\pi f$$

For the high-pass filter, the used components values were $R = 22k \text{ ohm}$, $C = 1.5nF$ using cut-off frequency formulas $f_{3dB} = \frac{1}{2\pi RC} = \frac{1}{2\pi(2200) \times 1.5 \times 10^{-9}}$, $f_{3dB} = 4822.9Hz$ This is theoretical cutoff frequency.

At this frequency, the output voltage is reduced by $\frac{1}{\sqrt{2}}$. The $V_{in} = 5.12$ The expected output at the cutoff frequency is $V_{out} = \frac{5.12}{\sqrt{2}} = 3.62$

In the measured data at 5000Hz, the V_{out} voltage was $V_{out} = 3.72V$.

This is very close to the theoretical value of 3.62V, meaning that the measured cut-off frequency is close to 5000Hz. Therefore, the experimental cut-off frequency agrees well with the theoretical value.

Next, the attenuation was calculated using $A_{dB} = 20 \log \left(\frac{V_{out}}{V_{in}} \right)$. At 200 Hz $A_{dB} = 20 \log \left(\frac{0.232}{5.12} \right)$

$A_{dB} = -26.87 \text{ dB}$ large negative value confirms that low frequencies are strongly attenuated.

Now at 100000Hz $A_{dB} = 20 \log \left(\frac{5.08}{5.12} \right) = 0.068 \text{ dB}$ This value is very close to 0dB, meaning that the high-frequency signal passes through almost without attenuation.

The phase shift also follows the expected trend. At low frequencies, the phase shift was close to 90° . At 200Hz, the measured phase shift was 88.8° , which agrees with the theoretical prediction that the phase approaches 90° as the frequency approaches zero.

Near the cut-off frequency, the phase shift should be approximately 45° . At 5000Hz, the measured phase shift was 44.6° . This is extremely close to the theoretical value, it confirms that the cut-off frequency is located near 5000,Hz.

At high frequencies, the phase shift approached 0° . At 100000Hz, the measured phase shift was 2.88° , which is close to zero. This confirms that at high frequencies, the capacitor behaves almost like a short circuit, so the output signal becomes almost to the input signal.

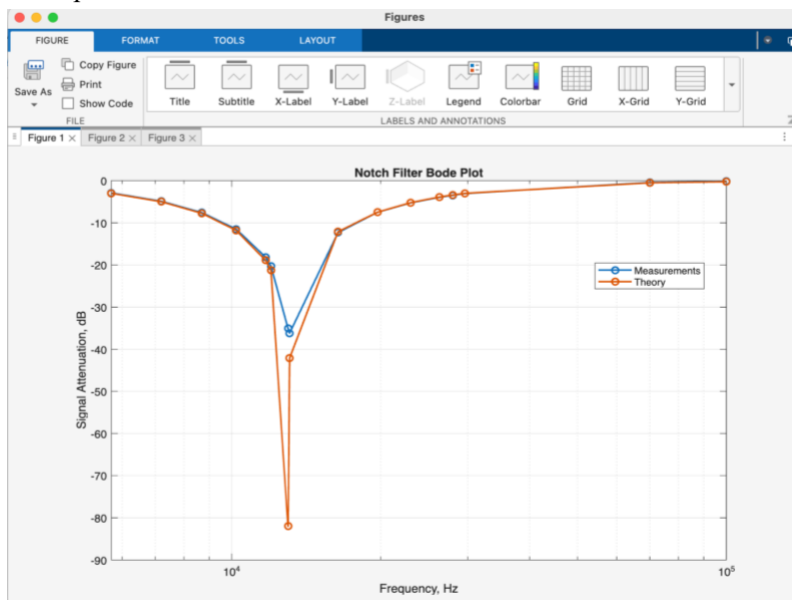
The measured data matches the theoretical values very closely. The output voltage was very small at low frequencies and almost equal to the input voltage at high frequencies. The phase shift also changed from approximately 90 at low frequencies to approximately 0 at high frequencies. Overall, the high-pass frequency behaved as expected, the measured cut-off frequency, attenuation, and phase shift all showed good agreement with theoretical values.

3.2 Notch- Evaluation

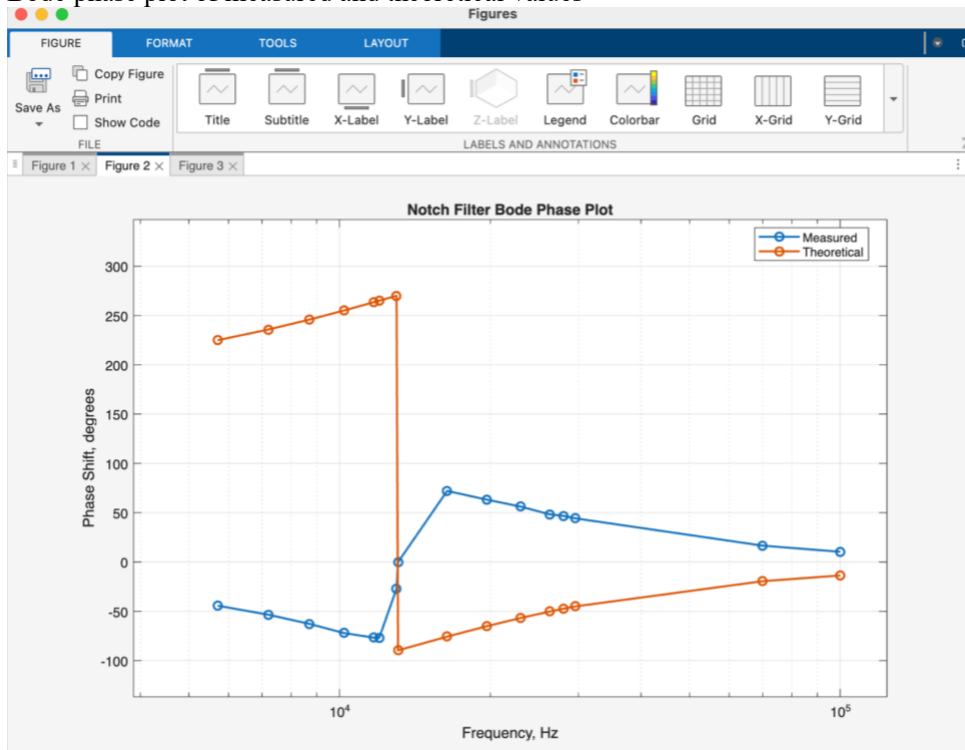
By amplification of the signal at each frequency using formula $|A| = \frac{1}{\sqrt{1 + \left(\frac{R}{\omega L - \frac{1}{\omega C}} \right)^2}}$ and $\varphi = \arctan \left(\frac{R}{\omega L - \frac{1}{\omega C}} \right)$

We obtain the theoretical values which are presented in the log scale and plotted alongside our measured values in previous figures

Bode plot of measured and theoretical values

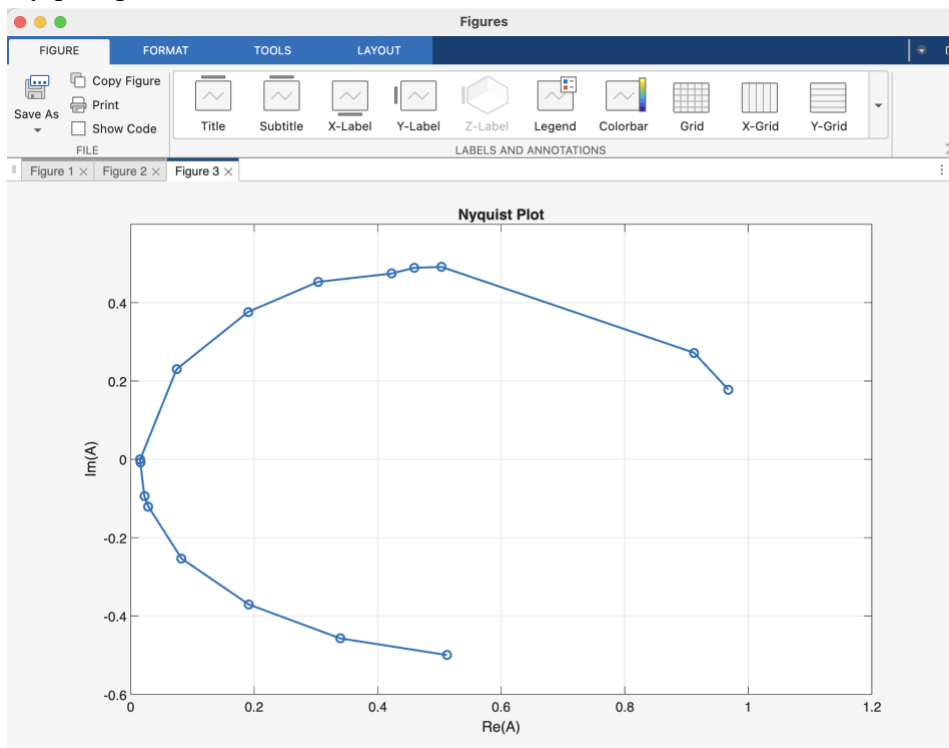


Bode phase plot of measured and theoretical values



Lastly, by taking the x-axis as $\frac{V_{out}}{V_{in}} \cos P$ and y-axis as $\frac{V_{out}}{V_{in}} \sin P$, where P is phase of measurements. The Nyquist plot of the imaginary parts vs real part of the transfer function as frequency changes is presented in this figure.

Nyquist plot of the measure values from the notch filter



4. Conclusion

In the first part we observed high-pass filter circuit impact of a to voltage amplitude and phase at the different frequencies. We confirmed that at higher frequencies above the cutoff the phase is less affected and the attenuation of the amplitude is 1, meaning the output is equal to input. The measured cut-off frequency was close to 5000Hz, which agrees well with the theoretical value of 4822.9Hz. The phase shift also behaved as expected, starting close to 90 at low frequencies and approaching 0 at high frequencies.

The experiment confirmed the theoretical behavior of passive filters. The high-pass filter attenuated low frequencies and allowed high frequencies to pass, while the notch filter strongly attenuated a narrow range of frequencies around center frequency. Small deviations between theoretical and measured values were most likely caused by component tolerances, internal resistance, oscilloscope uncertainty, and small measurement errors.

5. Reference

Uwe, P. (2024, 11 21). Retrieved from Uwe Pagel - Constructor University Natural Sciences Laboratory